

Recommended project size

Requirement	Recommended level
Core functionality	At least 4–10 meaningful functional features
User roles	At least 2 user roles
Database	approximately 3–8 related entities/tables
Authentication	Required
CRUD operations	Required where appropriate
API/backend	Required for medium/large projects
Testing	Testing
UI	Functional and consistent interface
Documentation	Requirements + design + architecture + testing
Version control	Git/GitHub/GitLab
Project management	Agile/Scrum recommended
Final presentation/demo	Required

1. System functionality

The project should have approximately **4–10 substantial user-oriented features**, for example:

For example, AI-Powered Academic Advisor

Possible features:

1. User registration/login
2. Student profile management
3. Curriculum management
4. Completed-course tracking
5. Prerequisite checking
6. Course recommendation
7. Personalized study-plan generation

8. Academic progress dashboard
9. AI conversational assistant
10. Administrator management

Recommended project architecture

Frontend → Backend/API → Database

For example:

Web Interface



REST API / Backend



Business Logic



Database



External API / AI service (if applicable)

A well-designed monolithic application is acceptable.

Non-functional requirements

You should also demonstrate that the system is not only functional but reasonably engineered.

I would require at least:

- **Usability** – understandable interface
- **Performance** – reasonable response time
- **Security** – authentication, authorization, input validation
- **Reliability** – error handling
- **Maintainability** – organized code and documentation

Suggested Agile requirement

- Product Backlog
- User Stories
- User Story estimation
- **7 iterations/sprints**
- Sprint Planning
- Sprint Review/Demo
- Sprint Retrospective
- Burndown/progress tracking
- Final working product

Deliverables

Stage	Deliverable
1. Project Proposal	Problem, objectives, target users, scope
2. Requirements	Functional/non-functional requirements, user stories
3. Product Backlog	Prioritized backlog + estimation
4. UI/UX & Architecture	Wireframes, ER diagram, architecture
5. Sprint 1–2	Basic system + authentication/database
6. Sprint 3–4	Core functionality
7. Sprint 5–6	Advanced functionality/integration
8. Sprint 7	Testing, bug fixing, finalization
9. Final Project	Working software
10. Final Documentation	Technical report + user guide
11. Presentation	Presentation + live demonstration